

Anonymous ARR Rebuttal

Anonymous

1 Additional materials

Model	COH	SYN	PHR	VOC	GRA	CON	Avg.
ModernBERT (Ours)	0.65	0.68	0.66	0.68	0.68	0.70	0.67
BERT [1]	0.57	0.66	0.65	0.63	0.63	0.66	0.63
RoBERTa [1]	0.62	0.67	0.69	0.65	0.71	0.71	0.67
MTL _{vanilla} [1]	0.63	0.69	0.72	0.67	0.72	0.71	0.69
MTAA [1]	0.63	0.69	0.71	0.67	0.73	0.72	0.69

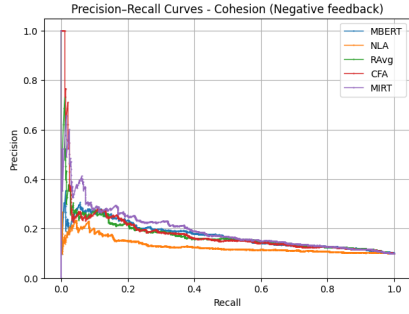
Table 1: Comparison of our ModernBERT-based grader against [1]’s graders in terms of Spearman’s rank coefficient against the original ELLIPSE test set labels.

Model	Analytic Dimension				
	CON	ORG	LAN	VOC	MCH
NLA	21.46	16.34	14.23	20.87	22.66
Q-RAvg	22.01	27.51	15.37	16.31	25.87
Q-WAvg	23.36	29.77	14.90	14.86	29.47
Q-CFA	21.94	24.27	15.35	14.52	17.84
Q-MIRT	20.90	28.44	16.11	18.20	19.12

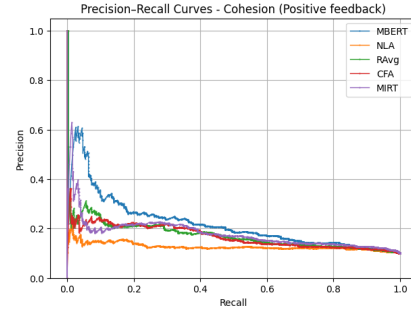
Table 2: Self-referential evaluation in terms of $F_{0.5}$ for **Negative** feedback (IC-NALE EE) using **Qwen 2.5 72B**.

Model	Analytic Dimension				
	CON	ORG	LAN	VOC	MCH
NLA	14.15	27.23	24.26	12.32	18.82
Q-RAvg	17.36	20.41	26.59	22.46	19.94
Q-WAvg	16.31	21.22	26.16	19.21	19.07
Q-CFA	14.17	20.64	24.75	18.03	18.67
Q-MIRT	19.93	19.43	23.52	19.86	14.86

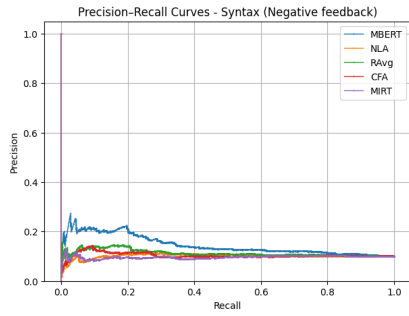
Table 3: Self-referential evaluation in terms of $F_{0.5}$ for **Positive** feedback (IC-NALE EE) using **Qwen 2.5 72B**.



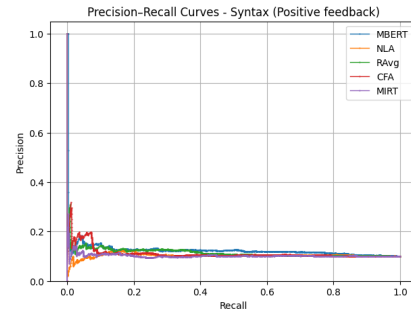
(a) Cohesion - negative feedback



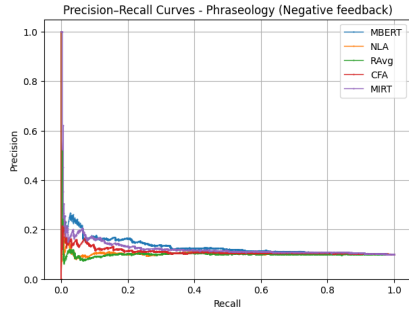
(b) Cohesion - positive feedback



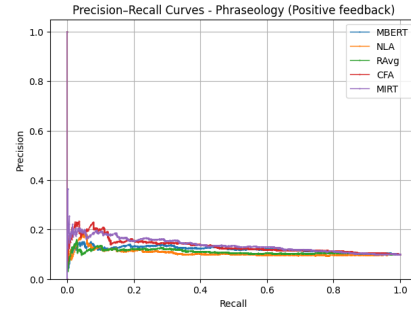
(c) Syntax - negative feedback



(d) Syntax - positive feedback

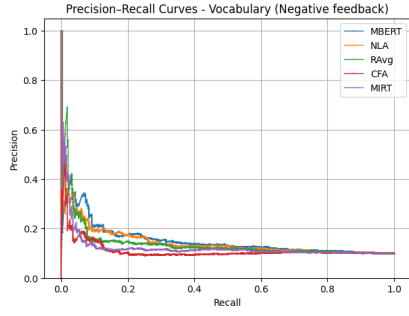


(e) Phraseology - negative feedback

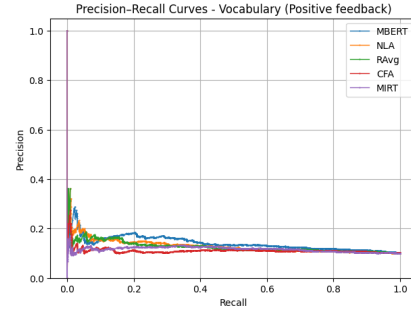


(f) Phraseology - positive feedback

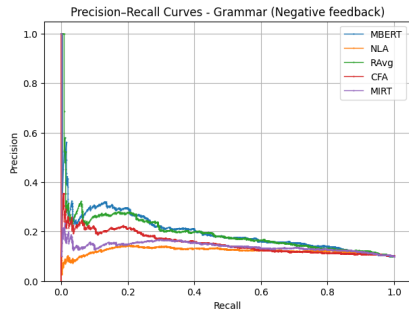
Figure 1: Precision-recall curves for self-referential evaluation results on EL-LIPSE using GPT-4.1-mini (1 of 2).



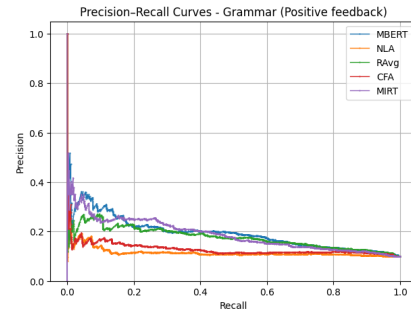
(a) Vocabulary - negative feedback



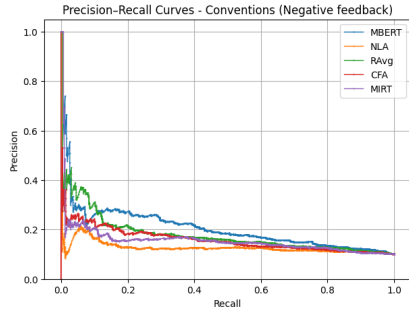
(b) Vocabulary - positive feedback



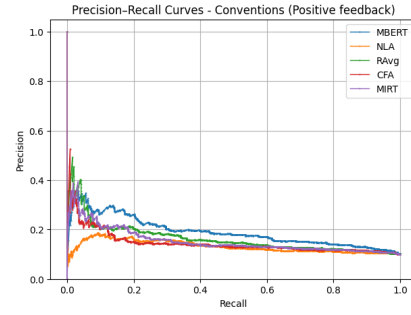
(c) Grammar - negative feedback



(d) Grammar - positive feedback



(e) Conventions - negative feedback



(f) Conventions - positive feedback

Figure 2: Precision-recall curves for self-referential evaluation results on EL-LIPSE using GPT-4.1-mini (2 of 2).

References

- [1] S. Chen, Y. Lan, and Z. Yuan, “A multi-task automated assessment system for essay scoring,” in *International Conference on Artificial Intelligence in Education*, pp. 276–283, Springer, 2024.